🧠 Task Manager (Python CLI)

📌 Description

The Task Manager is a command-line application that allows users to create, manage, and track tasks efficiently. It supports features such as task prioritization, due dates, tagging, and status updates.

🚀 Features

- Create and manage tasks
- Set task priorities (Low, Medium, High, Urgent)
- Assign due dates
- Add and remove tags
- Update task status (todo, in_progress, review, done)
- Filter tasks by status, priority, or overdue
- View task statistics
- Persistent storage using local file system

🛠️ Technologies Used

- Python 3.11+
- Built-in libraries:
  - argparse (CLI handling)
  - datetime (date management)
- JSON (for storage)

📦 Installation

Prerequisites

- Python 3.11 or higher

Steps

```
git clone <repository-url>
```

cd task-manager

No additional installation required.

▶️ Usage

Create a Task

python cli.py create "Task Title" --description "Task description" --priority 2 --due "2024-01-31" --tags "tag1,tag2"

List Tasks

python cli.py list

python cli.py list --status todo

python cli.py list --priority 3

python cli.py list --overdue

Update Tasks

python cli.py update-status <task_id> <new_status>

python cli.py update-priority <task_id> <new_priority>

python cli.py update-due-date <task_id> "YYYY-MM-DD"

Manage Tags

python cli.py add-tag <task_id> "tag"

python cli.py remove-tag <task_id> "tag"

View Details

python cli.py show <task_id>

python cli.py stats

⚙️ Configuration

- Tasks are stored in a local JSON file (tasks.json)
- No external configuration required

🧩 Code Structure

```
use-cases/code-algorithms/TaskManager/
├── cli.py          # Entry point (CLI interface)
├── task_manager.py    # Core business logic
├── models.py        # Task data models & enums
├── storage.py       # Data persistence layer
├── task_parser.py     # Text parsing logic
├── task_priority.py   # Priority handling
└── tests/          # Unit tests
```

⚠️ Troubleshooting

- **Command not found**
  → Ensure Python is installed and in PATH
- **Invalid date format**
  → Use format: YYYY-MM-DD
- **Task not found**
  → Check task ID using list command

🤝 Contributing

1. Fork the repository
2. Create a new branch
3. Make your changes
4. Submit a pull request

📄 License

This project is for educational purposes.